Lesson Notes

MODULE 1 | LESSON 4

THE MATHEMATICS OF CLASSICAL PORTFOLIO THEORY

Reading Time	120 minutes
Prior Knowledge	None
Keywords	Portfolio frontier, Efficient frontier, Minimum global variance portfolio, Tangency portfolio, Language of quadratic programming, Formulating a mean-variance optimization problem in the language of QP.

The previous lesson showed us how to go from utility theory to propose a program to maximize an investor's utility through a mean-variance optimization problem. In this lesson, we introduce the mathematics required to solve the optimization problem and look at a few of the nuances of the topic.

1. Finding the Portfolio Frontier

In this lesson, we will look at how to compute the efficient frontier, the global minimum variance portfolio, and the most efficient portfolio and how to deal with constraints.

Given a set of N assets, we choose to characterize a portfolio by the vector of its weights, i.e., $w^T = (w_1, w_2, \ldots, w_N)$. Recall that for us, a vector is a column vector. Hence, in our setup, w is an $N \times 1$ column vector, w^T, the transpose of w is a $1 \times N$ row vector, instead.

A portfolio is a **frontier portfolio** if it has the minimum variance among portfolios that have the same expected rate of return, that is if and only if the portfolio weights w^T are the solution of the following quadratic program:

$$\begin{aligned} &\min_w \frac{1}{2} w^T \Sigma w \\ &\text{subject to} \quad w^T r = E[r_p] \quad \text{and} \\ &\qquad\qquad\quad w^T \iota = 1, \end{aligned}$$

with Σ the $N \times N$ covariance matrix of the assets returns, r the $N \times 1$ vector of returns, ι, the $N \times 1$ vector of ones, i.e., $\iota^T = (1, 1, \ldots, 1)$, and $E[r_p]$ the *desired* expected rate of return on the portfolio.

Putting a $\frac{1}{2}$ in front of the variance of the portfolio is only for mathematical convenience and does not alter the optimization.

While we will not insist on the full derivation of all formulae, it is useful for a student to see how certain problems are solved. A quadratic program like the one above, since it does not involve any inequality constraints, can be solved using the traditional methods of calculus, e.g., the method of the Lagrange multipliers. The Lagrangian function is defined as

$$L(w, \lambda_1, \lambda_2) := \frac{1}{2} w^T \Sigma w + \lambda_1 (E[r_p] - w^T r) + \lambda_2 (1 - w^T \iota).$$

with $\lambda_1 > 0$ and $\lambda_2 > 0$ and which can be thought as the penalties for not satisfying the constraints. Now, the problem can be solved in the usual way, i.e. using multivariable calculus for $L(\cdot)$, i.e.:

$$\min_{w, \lambda_1, \lambda_2} \frac{1}{2} w^T \Sigma w + \lambda_1 (E[r_p] - w^T r) + \lambda_2 (1 - w^T \iota).$$

Students who have not had much exposure to these types of problems should pay attention to the fact that $L(\cdot)$ is a function of $N+2$ variables since w is a vector of N elements. Deriving the first order conditions would be very messy when N is large, and this is where differential calculus for matrices comes in handy. We will not provide a full theory of this and will limit ourselves to illustrating the relevant formulas that are found useful.

2. A Basic Introduction to Differential Calculus in Matrix Form

This is a very brief section and will only touch on very basic facts needed for our development, but students who are interested in this topic should study it. Matrix differential calculus is also an important tool of advanced econometrics.

We can think of $w^T\Sigma w$ as a function from $R^N \mapsto R$ defined by $f : w \mapsto f(w) := w^T\Sigma w$.

If we take the first derivative with respect to the vector w, this amounts to doing the following:

$$\frac{\partial w^T\Sigma w}{\partial w} = \begin{pmatrix} \frac{\partial w^T\Sigma w}{\partial w_1} \\ \frac{\partial w^T\Sigma w}{\partial w_2} \\ \vdots \\ \frac{\partial w^T\Sigma w}{\partial w_N} \end{pmatrix} = 2\Sigma w.$$

The proof is not difficult, albeit tedious, but the result is very simple; the language of matrices is powerful. We also have two other multivariable terms to derive $w^T r$ and $w^T \iota$ and which are the same type of function of w. This time, we consider the function $g : R^N \mapsto R$ defined as $g(w) = w^T r = \sum_{i=1}^N w_i r_i$. Then

$$\frac{\partial w^T r}{\partial w} = \begin{pmatrix} \frac{\partial w^T r}{\partial w_1} \\ \frac{\partial w^T r}{\partial w_2} \\ \vdots \\ \frac{\partial w^T r}{\partial w_N} \end{pmatrix} = r.$$

Using exactly the same steps, then we also find that $\frac{\partial w^T \iota}{\partial w} = \iota$.

3. Back to the Efficient Frontier Task

Using the results just found, we can now derive the first order conditions for the Lagrangian L :

$$\frac{\partial L}{\partial w} = \Sigma w - \lambda_1 r - \lambda_2 \iota = \mathbf{0}$$
$$\frac{\partial L}{\partial \lambda_1} = E[r_p] - w^T r = 0$$
$$\frac{\partial L}{\partial \lambda_2} = 1 - w^T \iota = 0,$$

with $\mathbf{0}$ an $N \times 1$ vector of zeroes. A very simple and elegant finding. In particular, since Σ is a positive definite matrix (variance-covariance matrices are!), the first order conditions are not only necessary, but also sufficient to find the global minimum.

Now we need to solve the system of equations we derived. From the first equation, we get

$$w = \lambda_1 \Sigma^{-1} r + \lambda_2 \Sigma^{-1} \iota.$$

If we multiply both the right and the left sides of this equation by r^T and use the second equation from the first order conditions, we get

$$E[r_p] = r^T w = \lambda_1 r^T \Sigma^{-1} r + \lambda_2 r^T \Sigma^{-1} \iota.$$

Doing the same, but this time multiplying by ι^T and using the third equation from the first order conditions, yields

$$1 = \iota^T w = \lambda_1 \iota^T \Sigma^{-1} r + \lambda_2 \iota^T \Sigma^{-1} \iota.$$

What have we achieved with this tedious algebra? Well, now we have a system of two linear equations:

$$\lambda_1 r^T\Sigma^{-1}r + \lambda_2 r^T\Sigma^{-1}\iota = E[r_p]$$
$$\lambda_1 \iota^T\Sigma^{-1}r + \lambda_2 \iota^T\Sigma^{-1}\iota = 1$$

which can be solved with respect to λ_1 and λ_2. The further (simple) algebraic manipulations will be spared and the solutions provided:

$$\lambda_1 = \frac{AE[r_p] - B}{D}$$
$$\lambda_2 = \frac{C - BE[r_p]}{D},$$

where

$$A = \iota^T\Sigma^{-1}\iota,\ B = \iota^T\Sigma^{-1}r = r^T\Sigma^{-1}\iota,\ C = r^T\Sigma^{-1}r,\ D = AC - B^2.$$

Finally, substituting these in the formula for w above, we get

$$w^\star = \frac{1}{D}\left(C\Sigma^{-1}\iota - B\Sigma^{-1}r\right) + \frac{1}{D}\left(A\Sigma^{-1}r - B\Sigma^{-1}\iota\right)E[r_p].$$

Notice that the optimal weight has the form:

$$w_p^\star = a + bE[r_p]$$

with

$$a = \frac{1}{D}\left(C\Sigma^{-1}\iota - B\Sigma^{-1}r\right), \quad \text{and} \quad b = \frac{1}{D}\left(A\Sigma^{-1}r - B\Sigma^{-1}\iota\right).$$

a and b are vectors, clearly. We use the subscript p to make the point that the optimal weights depend on the specific selected expected rate of return, $E[r_p]$.

We say that w_p^* is a **frontier portfolio** and the set of all frontier portfolios is called the **portfolio frontier**.

4. Two-Fund Separation

Let A and B be portfolios with expected return equal 0 and 1, respectively and let w_A and w_B be the weights that characterize these portfolios according to the formula we found in the previous section. Then, $w_A = a$ and $w_B = a + b.$

Now, we want a portfolio with expected return $E[r_q]$. Ideally that requires the weights $w = a + bE[r_q]$ which requires a lot of computations every time we change $E[r_q]$. However it is possible to obtain those weights without solving an optimization problem and just relying on portfolios A and B. More specifically, we invest $1 - E[r_q]$ in portfolio A and $E[r_q]$ in portfolio B. Note that this is legitimate as the sum of the weights of the new portfolio formed with A and B is equal to 1. Let's verify that this portfolio yields a return of $E[r_q]$. Let's call the newly formed portfolio P. Then

$$E[r_p] = (1 - E[r_q]) * r_A + E[r_q] \cdot r_B = (1 - E[r_q]) * 0 + E[r_q] \cdot 1 = E[r_q].$$

Recall that A is a portfolio with a return 0 and B a portfolio with a return of 1. So the portfolio does what we need it to do in terms of expected return.
The weights of the new portfolio can be found in terms of the weights of A and B as well:

$$w_p := w_A * (1 - E[r_q]) + w_B(E[r_q]) = a(1 - E[r_q]) + (a + b)E[r_q] = a + bE[r_q] = w_q,$$

just as we needed.

This fact allows us to state the following result:

Theorem (Two-fund separation). *Every portfolio on the portfolio frontier can be generated by any two distinct frontier portfolios not having the same expected return.*

To prove this theorem, it is enough to consider the weights w_{p_1} and w_{p_2} for two frontier portfolios such that $E[r_{[1}] \neq E[r_{p_2}]$ and let q be any other portfolio on the portfolio frontier. We form a new portfolio consisting of portfolio p_1 and portfolio p_2 whose weights are γ and $1-\gamma$. Then, using the earlier remark, we can write:

$$\begin{aligned}\gamma w_{p_1} + (1-\gamma)w_{p_2} &= \gamma(a + bE[r_{p_1}]) + (1-\gamma)(a + bE[r_{p_2}]) \\ &= a + b(\gamma E[r_{p_1}] + (1-\gamma)E[r_{p_2}]) = a + hE[r_q] = w_q.\end{aligned}$$

5. Global Minimum Variance Portfolio

Our next task is to identify the weights associated with the **global minimum variance portfolio**. This is actually simpler to do than computing the weights of a portfolio on the portfolio frontier since it is enough to solve the following program:

$$\min_w \frac{1}{2} w^T \Sigma w$$
$$\textbf{subject to} \quad w^t \iota = 1,$$

It is basically the same problem as before with the condition $w^T r = E[r_p]$ removed. After some algebra (students are invited to perform the calculations as an exercise) one gets:

$$w_{GMV} = \frac{\Sigma^{-1}\iota}{\iota^T \Sigma^{-1}\iota}.$$

The expected return on the global minimum variance portfolio is then given by

$$E[r_{GMV}] = w_{GMV}^T \times r = \frac{\iota^T \Sigma^{-1} r}{\iota^T \Sigma^{-1}\iota} = \frac{B}{A}.$$

If we compare the solution for the generic portfolio on the portfolio frontier, we see that the variance for the portfolio with $E[r_p]$ is greater than that of the global minimum variance portfolio when $E[r_p] > \frac{B}{A}$. Also, if we look at our formula for λ_1, the last inequality implies that the Lagrange multiplier for the mean return constraint is positive. The set of minimum variance portfolios that satisfy this condition is called the **mean-variance efficient set**.

When $E[r_p] < \frac{B}{A}$, then $\lambda_1 < 0$ which implies that a lower mean return is consistent with a higher variance (the return is less than that of the global minimum variance portfolio and the variance is higher than that of the global minimum variance portfolio). Hence, no economic agent who likes mean and dislikes variance would want to hold this portfolio.

The variance of the global minimum variance portfolio can be calculated using the formula:

$$w_{GMV}^T \Sigma w_{GMV} = \frac{\iota^T \Sigma^{1-} \Sigma \Sigma^{-1}\iota}{(\iota^T \Sigma^{-1}\iota)^2} = \frac{\iota^T \Sigma^{1}\iota}{(\iota^T \Sigma^{-1}\iota)^2} = \frac{1}{\iota^T \Sigma^{-1}\iota}.$$

Example (Naive diversification). What is the variance of a portfolio that assigns the same weight to each of the N assets? In this case:

$$w = \frac{1}{N}\iota.$$

and the variance of the portfolio is

$$w^T \Sigma w = \frac{1}{N^2}\iota^T \Sigma \iota.$$

If we assume that all assets have the same variance and correlation, then with a few algebraic manipulations:

$$\frac{1}{N^2}\iota^T \Sigma \iota = \frac{1}{N^2}\left(\sum_{i=1}^{N} \sigma^2 + \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} \sigma^2 \rho\right) = \sigma^2 \rho + \frac{1}{N}\sigma^2(1-\rho).$$

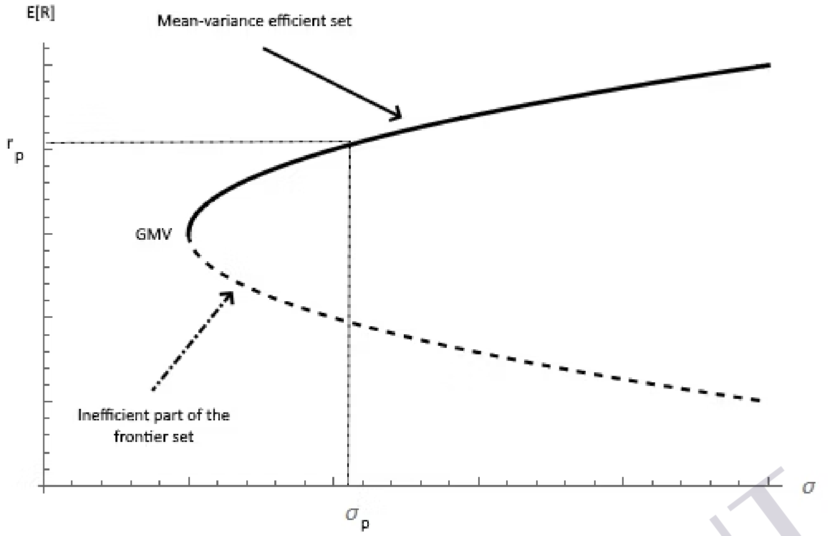
We know that the variance cannot be a negative number. This implies (the student is encouraged to do the algebra!) that

$$\rho > -\frac{1}{N-1}.$$

This is an interesting result: As the number of assets in the portfolio increases, it gets more and more difficult for the assets to be negatively correlated. Ideally, when $N \to \infty$, the correlation ρ must be nonnegative.

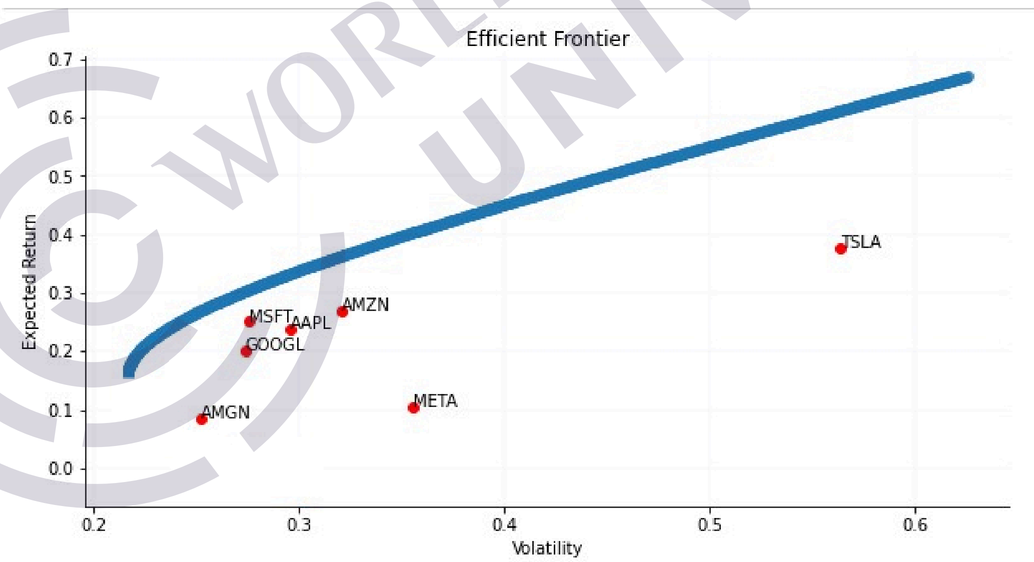
All our work can be summarized through the graph of Figure 1. Note that the dotted line represents the inefficient part of the frontier set.

Figure 1: The Mean-Variance Efficient Set and Global Minimum Variance Portfolio (GMV)



Example (Portfolio with 7 well known assets). We know enough to present a calculation with real assets. All the stock are well-known. We notice that at one side of the spectrum, Tesla has high return and high risk. On the other hand, Amgen, as a biotech/big pharma, has much less risk, but allows for a lower return. The efficient frontier for this example is exhibited in Figure 2.

Figure 2: A Computation of the Efficient Frontier Using a Group of Seven Assets (reference period Jan. 2, 2015–Jul. 7, 2022)



6. Adding a Riskless Asset and Tangency Portfolio

Until now, we considered all assets to be risky. Now we assume that there is a riskless asset. More specifically, we assume that we have N risky assets and 1 riskless asset. The risk-free rate is denoted by r_f. When a riskless asset is introduced, certain interesting developments take place.

Let p denote a frontier portfolio that consists of all $N+1$ assets, and use w_p for the $N \times 1$ vector of portfolio weights of p that relate just to the risky assets. Then, w_p can be obtained as the solution of the following quadratic program:

$$\min_{w_1, w_N} \frac{1}{2} w^T \Sigma w$$

$$\text{subject to } w^T r + (1 - w^T \iota) r_f = E[r_p].$$

Note that when $1 - w^T \iota < 0$ that is equivalent to borrowing money at the risk-free rate to buy more of the risky assets, and, as students already noticed, we remove the second constraint, i.e., the weights of the risky assets no longer need to sum up to 1. Like we did before, we form the Lagrangian, and the solution w_p can be obtained from the first order conditions associated with the Lagrangian function:

$$L(w_1, \dots w_N, \lambda) = \frac{1}{2} w^T \Sigma w + \lambda \big(E[r_p] - w^T r - (1 - w^T \iota) r_f \big).$$

The first order conditions are

$$\frac{\partial L(w_1, \dots, w_N, \lambda)}{\partial w} = \Sigma w + \lambda(-r + r_f \iota) = 0$$
$$\frac{\partial L(w_1, \dots, w_N, \lambda)}{\partial \lambda} = E[r_p] - w^T r - (1 - w^T \iota) r_f = 0.$$

Therefore, if w_p, λ are the solutions of the quadratic programming, w_p must satisfy the following system of equations:

$$\Sigma w_p = \lambda (r - r_f \iota), \quad r_f + w_p^T (r - r_f \iota) = E[r_p].$$

The system can be solved, and it yields

$$\lambda = \frac{E[r_p] - r_f}{E},$$
$$w_p = \lambda \Sigma^{-1} (r - r_f \iota) = \Sigma^{-1} (r - r_f \iota) \frac{E[r_p] - r_f}{E},$$

with $E = (r - r_f \iota)^T \Sigma^{-1} (r - r_f \iota)$. The scalar quantity E can be rewritten using our original quantities A, B, C, i.e.:

$$E = C - 2 B r_f + A r_f^2.$$

Since $B^2 - AC < 0$ and $A > 0$ (it is a positive definite quadratic form because Σ is positive definite and therefore so is Σ^{-1} is too), the parabola that is associated with E is always above the x-axis, and thus, E is always positive. Then, the variance of the minimized portfolio is

$$\sigma_p^2 = w_p^T \Sigma w_p = \lambda^2 (r - r_f \iota)^T \Sigma^{-1} \Sigma \Sigma^{-1} (r - r_f \iota) = \lambda^2 E$$

and, substituting the formula found above for λ, finally yields

$$\sigma_p^2 = \frac{(E[r_p] - r_f)^2}{E}.$$

This implies that $|E[r_p] - r_f| = \sigma_p \sqrt{E}$ and that

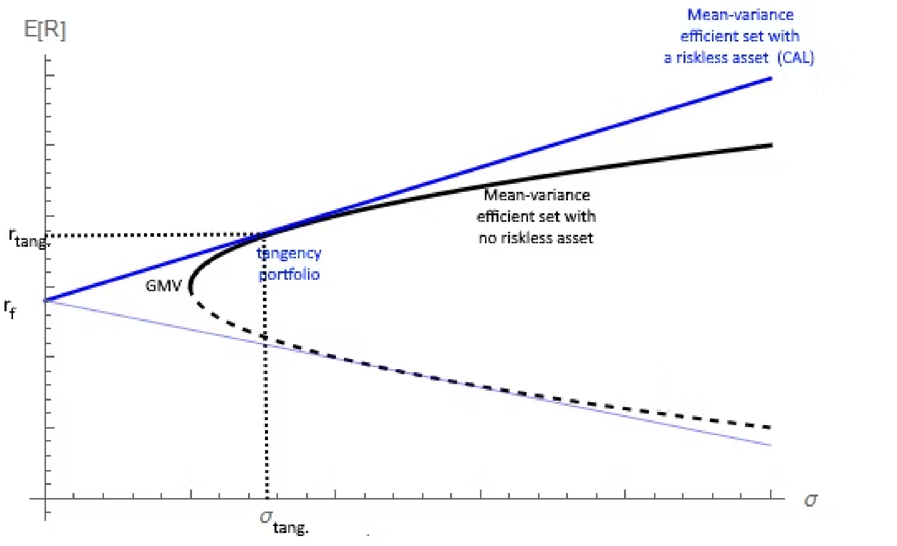
$$\sigma_p = \begin{cases} \frac{E[r_p] - r_f}{\sqrt{E}} & \text{if } E[r_p] \ge r_f; \\ -\frac{E[r_p] - r_f}{\sqrt{E}} & \text{if } E[r_p] < r_f. \end{cases}$$

or, equivalently,

$$E[r_p] = \begin{cases} r_f + \sqrt{E} \sigma_p & \text{if } E[r_p] \ge r_f; \\ r_f - \sqrt{E} \sigma_p & \text{if } E[r_p] < r_f. \end{cases}$$

This result illustrates an important consequence: When we add a riskless asset and we allow borrowing funds at that rate, then the portfolio frontier becomes two half-lines that cross at the point $(0, r_f)$ in the $(\sigma, E[r])$ space.

Figure 3: Efficient Frontier and Tangency Portfolio with and without a Riskless Asset



The upper branch of the V-shaped curve is a tangent line from r_f on the vertical axis to the mean-variance efficient set obtainable if one can only hold the risky assets and not the riskless asset. This upper branch is also known as the capital allocation line (CAL). Graphically, it appears as if there is only one portfolio on the CAL—and that contains only risky assets—and, as such, is also on the mean-variance efficient frontier constructed with risky assets only. This portfolio is known as the **tangency portfolio**. This can be observed in Figure 3.

One additional important feature of this portfolio is that the slope of such a portfolio is the same as that on the CAL line. We know that the slope of the CAL line is the Sharpe ratio.

There is one final question to address: how do we determine the tangency portfolio? The tangency portfolio is the solution of the following optimization problem:

$$\max_{w_1,...,w_N} \frac{w^T r - r_f}{\sqrt{w^T \Sigma w}}$$
$$\text{subject to } w^T \iota = 1.$$

This is not an easy problem, although it can be solved using numerical algorithms.

If we assume that $w^T r - r_f > 0$, which is a reasonable assumption because all it says is that our portfolio is able to beat the risk-free rate, one can show (it is not straightforward, but not entirely difficult either) that the program above is equivalent to the following one:

$$\max_{v_1,...,v_N} \frac{1}{\sqrt{v^T \Sigma v}}$$
$$\text{subject to } w^T \tilde{r} = 1.$$

with $\tilde{r} = r - r_f\iota$, or $\tilde{r}^T = (r_1 - r_f, r_2 - r_f, \ldots, r_N - r_f)$. Then, we can solve the problem to determine v as

$$\min_{v_1,...,v_N} \frac{1}{2} v^T \Sigma v$$
$$\text{subject to } v^T \tilde{r} = 1,$$

which is a standard quadratic programming. Using the method of the Lagrange multipliers once again, one finds (students should do the calculations as in this case they are actually easy)

$$v = \frac{\Sigma^{-1}\tilde{r}}{\tilde{r}^T \Sigma^{-1} \tilde{r}}.$$

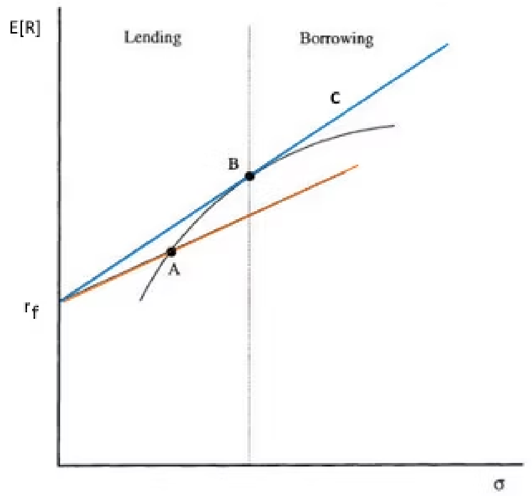
Once one has v, the weights for the tangency portfolio can be computed simply as

$$w_{TANG} = \frac{v}{v^T \iota} = \frac{\Sigma^{-1}(r - r_f\iota)}{B - r_f A}.$$

Where A and B are the same symbols used before.

The relevance of the tangency portfolio is due to the following consideration: In the presence of a riskless asset, *Tobin's two-fund theorem* simplifies because one of the two mutual funds is the riskless asset and the other is the tangency portfolio that contains only risky assets. In addition, this tells us one more thing: Different agents/investors may have a different appetite for risk (based on individual risk aversion); nevertheless, all investors should hold risky assets in the same proportion. The more conservative agents will dilute the risky assets by holding more of the risk-free asset, while aggressive agents may even borrow cash (at the risk-free rate) to increase their returns.

Figure 4: Different Portfolios for Different Investors



In figure 4, aggressive investors will actually borrow money and hold portfolio C. Investors who neither borrow nor lend money will hold the tangency portfolio B. The line that goes via portfolio A represents all combinations of risk-free lending and borrowing with portfolio A. These are not optimal combinations. An agent with zero appetite for risk will invest everything in a money market portfolio and earn the risk-free interest rate r_f.

Example (Putting the concepts to work). Assume that we have three risky assets whose expected returns are $0.05, 0.07$, and 0.11, respectively. Their individual volatilities are $0.10, 0.08$, and 0.15. The correlation between asset 1 and asset 2 is 0.4; the correlation between asset 1 and 3 is 0.5; and the correlation between asset 2 and 3 is -0.3. Finally, the risk-free interest rate is 0.01. What is the tangency portfolio?

First we need to compute the variance covariance matrix for these three assets. Students should check that it is:

$$\Sigma = \begin{pmatrix} 0.01 & 0.0032 & 0.0075 \\ 0.0032 & 0.0064 & -0.0036 \\ 0.0075 & -0.0036 & 0.0225 \end{pmatrix}$$

The vector $\tilde{r}^T = (r - r_f)^T = (0.04, 0.06, 0.10)$, instead. Then, we can find the weights for the portfolio v using the formula:

$$v = \frac{\Sigma^{-1}\tilde{r}}{\tilde{r}^T\Sigma^{-1}\tilde{r}}.$$

In order to do this we need first to invert the matrix Σ, which yields:

$$\Sigma^{-1} = \begin{pmatrix} 239.474 & -180.921 & -108.772 \\ -180.921 & 308.388 & 109.649 \\ -108.772 & 109.649 & 98.2456 \end{pmatrix}$$

and, after some algebra: $v^T = (-5.91986, 10.8287, 5.87073)$. Hence

$$w^T_{TANG} = (-0.549175, 1.00456, 0.544617).$$

The return and the volatility of the tangency portfolio are:

$$r_{TANG} = 0.1028, \quad \sigma_{TANG} = 0.0647.$$

Let's say that an investor would like to achieve a return of 15% instead of just 10.65%. How can they achieve that? Clearly, they must be willing to accept more risk, but the mechanics are pretty simple and use a linear combination of the risk-free asset and the tangency portfolio; that is, we need to find α such that:

$$\alpha r_f + (1-\alpha) r_{TANG} = 0.15.$$

Using $r_f = 0.01$ and $r_{TANG} = 0.1028$, and solving the simple linear equation, we get:

$$\alpha = -0.5086,$$

i.e., we are borrowing at the risk-free rate (that we had to borrow it was clear since we are aiming for an expected return higher than that of the tangency portfolio) and using the additional funds to buy more of the tangent portfolio.

The weights for the portfolio consisting of the risk-free asset and the tangent portfolio are $(-0.5086, 1.5086)$.

To implement this portfolio, we need to specify the weights in terms of risk-free asset and the original three assets:

$$(-0.5086, 1.5086 \times (-0.549175, 1.00456, 0.544617)) = (-0.5086, -0.8285, 1.5155, 0.8216).$$

One can check that this portfolio allows us to achieve an expected return of 15% as desired. The volatility for the new portfolio is 9.79%, higher than that for the tangency portfolio.

7. Quadratic Programming (QP)

Classical portfolio theory lends itself to a certain amount of criticism, yet it is useful and addresses specific real life problems faced by practitioners in their profession. What was done in the previous sections is all good and elegant, but it is not enough to solve all problems.

In what we did so far, for instance, short-selling is allowed, we did not assign any limit to the weights of each individual stock in the portfolio, we did not consider the cost of rebalancing portfolios, etc. Whenever we introduce inequality constraints (like when we say that no individual stock can represent more than, say 10% of the overall portfolio, that is akin to stating that $w_i \leq 0.10$ for each $i = 1, 2, \ldots, N$. This introduces inequality constraints.

When inequality constraints are introduced, closed form solutions are no longer possible, not in general at least.

Thankfully, almost any high-level programming language makes available packages to perform quadratic programming.

In its most general form, a quadratic programming looks like this:

$$\begin{aligned} \min_x \frac{1}{2} x^T P x + q^T x \\ \text{subject to} \quad Gx = g \\ Hx \leq h \end{aligned}$$

where x is, say an $n \times 1$ vector with respect to that which we are trying to optimize. P is a positive definite matrix, q a vector, and G, H are suitable matrices to write the constraints. Note that the first set of constraints are equality constraints; the second set instead consists of inequality constraints. Furthermore, the inequality is to be understood as being applied element-wise to the vectors Hx and h.

If Python is the programming language of choice, the (CVXOPT) package contains a routine to do QP. So does CVXPY (CVXPY). There are more! Regardless of what software package is being used, one has to pay attention to whether the particular routine is using $\max$ or $\min$ since different packages may have different setups. However, this is not an issue at all since

$$\min_x \frac{1}{2} x^T P x + q^T x = \max_x - \left(\frac{1}{2} x^T P x + q^T x \right),$$

but, we stress it, one needs to be aware of how one's software of choice works.

Example (Classical problem for the derivation of the efficient frontier). In what follows, to keep the notation simple, we will consider only three stocks, i.e., $w^T = (w_1, w_2, w_3)$, $\iota^T = (1, 1, 1)$, $r^T = (r_1, r_2, r_3)$.

The setup we used earlier was the following:

$$\min_w \frac{1}{2} w^T \Sigma w$$
$$\text{subject to} \quad w^T r = E[r_p] \quad \text{and}$$
$$w^T \iota = 1.$$

The first thing that we notice is that here we have no inequality constraints. Hence:

$$P = \Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 \end{pmatrix}, \quad q = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad G = \begin{pmatrix} r_1 & r_2 & r_3 \\ 1 & 1 & 1 \end{pmatrix}, \quad g = \begin{pmatrix} E[r_p] \\ 1 \end{pmatrix}$$

and we do not need any H matrix and h vector. However, some software (like the one used in the programming language R) write the matrices H and G as a single matrix and then one needs to tell the program how many rows pertain to equality constraints while the remaining ones are taken as inequality constraints.

Example (Portfolio optimization with limited max exposure to individual stocks). We can consider the same problem as above, i.e., short selling is allowed but we do not want the weight of any stock to exceed $0.40.$ Then, the problem becomes: stocks, i.e., $w^T = (w_1, w_2, w_3)$, $\iota^T = (1, 1, 1)$, $r^T = (r_1, r_2, r_3)$. The, setup we used earlier was the following:

$$\min_w \tfrac{1}{2} w^T \Sigma w$$
$$\text{subject to} \quad w^T r = E[r_p] \quad \text{and}$$
$$w^T \iota = 1$$
$$w_1 \leq 0.4$$
$$w_2 \leq 0.4$$
$$w_3 \leq 0.4.$$

In this case, the matrices P, G are as before and so are the vectors q and g. However, now we have:

$$H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad h = \begin{pmatrix} 0.4 \\ 0.4 \\ 0.4 \end{pmatrix}.$$

Some clients have strong environmental, social, and governance (ESG) feelings and may require that their portfolio managers do not invest in certain classes of assets. In that case, weights for assets that are forbidden by clients would be set to zero.

Example (Portfolio optimization with limited max and min exposure to individual stocks (box constraints)). We can consider the same problem as above, i.e., short selling is allowed, but we do not want the weight of any stock to exceed 0.40 and, in addition, the client has requested that the weight of the third asset does not go below 0.2 Then, the problem becomes:

$$\min_w \tfrac{1}{2} w^T \Sigma w$$
$$\text{subject to} \quad w^T r = E[r_p] \quad \text{and}$$
$$w^T \iota = 1$$
$$w_1 \leq 0.4$$
$$w_2 \leq 0.4$$
$$w_3 \leq 0.4$$
$$w_3 \geq 0.2.$$

The last inequality can be written as $\geq$ by simply using it as $-w_3 \leq -0.2.$ Hence, the matrices P, q, G, g remain as usual, but

$$H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix}, \quad h = \begin{pmatrix} 0.4 \\ 0.4 \\ 0.4 \\ -0.2 \end{pmatrix}.$$

Now we look at a more concrete example with specific names from everyday life.

Example (Application to the portfolio consisting of AAPL, MSFT, AMZN, GOOGL, META, TSLA, AMGN). We have already seen this setup in deriving the portfolio frontier. Let's say that we want to find the optimal portfolio that yields 35 return with no short selling constraints, but the sum of the weights needs to be 1. Then (using daily data from Jan. 2, 2015, to Jan. 15, 2022 to estimate returns and parameters), the optimal solution is given by the following weights:

```
0.22251740

0.34298749

0.44328915

0.13602145

−0.16611600

0.11255463

−0.09125412
```

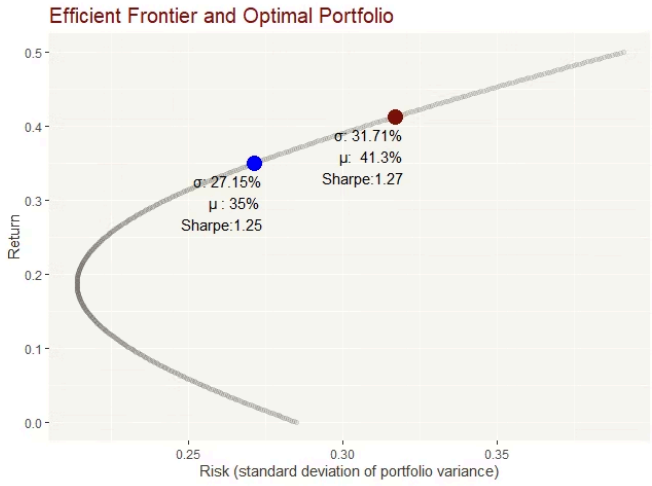
and the portfolio achieves a volatility of 0.2695.

Let's say now that we want the same return, but we do not want to be more than 0.05 in TSLA. Then, running a new quadratic program with an added constraint, we get

```
0.2537489

0.3877040

0.4809173

0.1310326

−0.1730621

0.0500000

−0.1303407
```

The computed volatility is now 0.2715, higher than before and which was to be expected as we added an additional constraint. The Sharpe ratio for this portfolio is 1.25 (see figure 5).

Figure 5: Required Solution (blue dot) vs. the Best Portfolio (red dot)



Looking at the portfolio frontier can be sort of fun, and it can look appealing, but ultimately, what really matters is the composition of the portfolio we need to implement for an institution or for individual clients. Still, one can produce plots like figure 5 by running the program for multiple values of desired expected returns.

To complete this section, we look at an interesting problem. What happens if lending and borrowing rates are not the same? Assume that r_l is the risk-free lending rate, and r_b is the borrowing rate. Then, we can define a new interest rate that depends on the particular situation, i.e., it is a function of the weights of the risky assets in the portfolio.

$$r(w) = \begin{cases} r_l & \text{if } w^T\iota - 1 \leq 0; \\ r_b & \text{if } w^T\iota - 1 \geq 0. \end{cases}$$

Using $r(w)$, the expected return for the portfolio can be written as:

$$E[r_p] = w^T r + (1 - w^t\iota)r(w).$$

and the portfolio frontier can be found by solving the programming problem:

$$\min_{w_1,...,w_N} \frac{1}{2} w^T \Sigma w$$
$$\text{subject to } \ w^T r + (1 - w^T\iota)r(w) = E[r_p].$$

This is NOT a quadratic program as the function $r(w)$ is not smooth (it is not differentiable everywhere).

To solve this problem, one has to actually solve two problems:

1. If $w^T\iota > 1$:

$$\min_{w_1,...,w_N} \frac{1}{2} w^T \Sigma w$$
$$\text{subject to } \ w^T r + (1 - w^T\iota)r_b = E[r_p].$$

1. and if $w^T\iota \leq 1$, then:

$$\min_{w_1,...,w_N} \frac{1}{2} w^T \Sigma w$$
$$\text{subject to } \ w^T r + (1 - w^T\iota)r_l = E[r_p].$$

Once the two quadratic programs are solved, we obtain two systems of weights. The solution (portfolio) with the lower variance is the solution we want. QP is used for each of the two sub-problems though.

We conclude this section with some Python code where the CVXPY package is used to implement quadratic programming and determine the weights for an optimal portfolio when both equality and inequality constraints are used. Generally speaking, from a didactic point of view, it is much better when students write their own code. But as this skill is an important part of this module, we would like to make sure that everyone gains this skill. Of course, it is possible to insert more complex constraints (box constraints), but that requires larger matrices to set up the problem. Make sure that you now how to use pip install if your Anaconda is missing certain libraries.

```
import pandas as pd
import numpy as np
import cvxpy as cvx
import yfinance as yf
from datetime import datetime

## define the stocks for our portfolio (via tickers) ###
assets = ["AAPL", "AMZN","TSLA","GOOGL","META","MSFT","AMGN", "V"]
# Define the start date
start = "2020-03-15"
# Today's date
today = datetime.today().strftime('%Y-%m-%d')
# Acquire data
df_prices = yf.download(assets,start=start_date,end=today)['Adj Close']

# We transform the data to its logarithmic returns
df = np.log(df_prices).diff()
# Drop the first row because we loose information we the logarithmic return.
df = df.dropna()  # this is a dataframe

# Compute the Expected Returns, we multiply daily return by 252
#because there are 252 business days in the US.
exp_returns = df.mean()*252
cova = df.cov()*252

m=exp_returns.shape[0]

# WANT: optimal portfolio with the following constraints
# 1. no short sales are allowed  that is the weights for all stocks are >=0
# 2. the maximum weight allowed for Amazon is 0.2
# 3. the maxium weight allowed for Tesla is 0.3
# 4. the sum of the weights is 1
# 5. We need to have a weight of 0.05 for Microsoft

#def optimize_portfolio(exp_returns, index_weights, scale=.0000):
Q = np.array(cova)   # Q is variance-covariance matrix of assets
q = np.array(exp_returns) # q is the exptected returns of assets
#   Inequalities are written as Gx <= h
G = np.array(np.matrix([[-1,0,0,0,0,0,0,0],[0,-1,0,0,0,0,0,0], \
                        [0,0,-1,0,0,0,0,0],[0,0,0,-1,0,0,0,0], \
                        [0,0,0,0,-1,0,0,0],[0,0,0,0,0,-1,0,0], \
                        [0,0,0,0,0,0,-1,0],[0,0,0,0,0,0,0,-1], \
                        [0,1,0,0,0,0,0,0],[0,0,1,0,0,0,0,0]]))
h =np.array([0,0,0,0,0,0,0,0,0.2,0.3])
# Equality constraints are written as Ax == b
A = np.array(np.matrix([[1,1,1,1,1,1,1,1],[0,0,0,0,0,1,0,0]]))
                # matrix A is used for the equality constraints
```

```
b = np.array([1, 0.05])
#
# Define and solve the quadratic problem using CVXPY:
# =====================================================
x = cvx.Variable(m)
prob = cvx.Problem(cvx.Maximize((-1/2)*cvx.quad_form(x, Q) + q.T @ x),
             [G @ x <= h,
              A @ x == b])
prob.solve()
#retrieve the weights of the optimized portfolio
x_values = x.value
#Print the optimal weights for each asset in the portfolio.
for i in range(m):
        print(assets[i], f"{x_values[i]: f}")

opt_ptfolio_ret = np.dot(x_values, exp_returns)
print("Expected return optimal porfolio: ", f"{opt_ptfolio_ret:.2%}")

opt_ptfolio_stdev = np.sqrt(np.dot(x_values, Q.dot(x_values.T)))
print("Expected std. deviation optimal porfolio: ", f"{opt_ptfolio_stdev:.2%}")
```

8. Conclusion

In this lesson, we learned about the nuances of classical mean-variance optimization to determine optimal portfolios. We covered the math necessary to derive the important formula, and we looked at a real-life application.

Behind the scenes, estimates for the mean of expected returns for individual assets and their estimated covariance are fed to an algorithm. In the next lesson, we will investigate more in depth the issues related to using estimates. We anticipate that it is not a small role, unfortunately. In particular, it is the estimated mean returns that can cause the most trouble. In an attempt to mitigate the issue, we will visit resampling techniques using both a simple bootstrap technique and the celebrated Michaud's resampling approach.